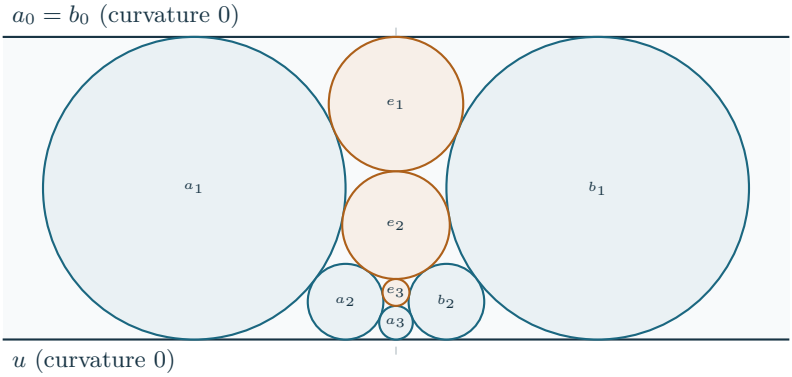
The exact seed for a uniform packing comparison

Reduced width $k = 4$: the two horizontal lines bound a strip of height 2. The orange disks e_1, e_2, e_3 form a tangent chain on the symmetry axis.



$$a_0 = b_0 = 0, \quad a_1 = b_1 = 1, \quad a_2 = b_2 = 4, \quad a_3 = b_3 = 9$$

$$(e_1, e_2, e_3) = \left(\frac{9}{4}, \frac{45}{16}, \frac{45}{4}\right)$$

a_1, b_1 and a_2, b_2 are distinct mirror disks with equal curvatures. $a_3 = b_3$ is the single shared axis disk; $a_0 = b_0$ is the upper line. Curvature is reciprocal radius; both marked lines have curvature 0.

The contact-height comparison at larger widths

Let $y_0 = 2$ and let y_j be the height of the contact below e_j. Define $z_j = \sqrt{1 - y_j/y_{j-1}}$; the endpoint symbols are $z_0 = z_k = 0$.

$$(1 + z_{j-1})(1 - z_j^2)(1 + z_{j+1}) = 1, \qquad z_0 = z_k = 0$$

$$k = 4 : \qquad (z_1, z_2, z_3) = \left(\frac{2}{3}, \frac{4}{5}, \frac{2}{3}\right)$$

The substitution $v_j = \log(1 + z_j)$ gives the convex comparison principle. This three-contact seed proves $z_j \geq 2/3$ for the reduced family $k \geq 4$.